

WILLIAM LAZCANO

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EDUCATION

B.S. Computer & Electrical Engineering — Dual Major

2022 – 2028 (Expected)

University of Nevada, Las Vegas | Coursework: Digital System Architecture, Data Structures, C++, Computer Systems, Circuit Analysis

TECHNICAL SKILLS

HDL: SystemVerilog, Verilog

Architecture: Custom ISA, ARM, MIPS, Multicycle & Single-Cycle processors

FPGA: Altera DE2-115, Quartus II, TimeQuest Timing Analyzer

Programming: C++, Python, Assembly (ARM/MIPS)

Tools: ModelSim, Git, Linux/Unix

Languages: Spanish (Native), English (Fluent)

ENGINEERING PROJECTS

Custom 8-Bit Multicycle Processor

Fall 2024

- Designed complete 8-bit CPU from scratch — custom 16-instruction ISA, FSM control unit, full datapath in SystemVerilog; no starter code
- Deployed on Altera DE2-115 FPGA executing 2x2 matrix operations at 50 MHz; verified all 16 instructions via ModelSim waveform analysis

ARM Processor with Extended ISA

Fall 2024

- Implemented 32-bit single-cycle ARM processor in Verilog; extended ALU from 2-bit to 3-bit control to add EOR (XOR) instruction
- Added LDRB with byte selector and zero-extension logic; timing analysis confirmed 55.8 MHz max frequency with positive slack

MIPS Processor with Subroutine Support

Fall 2024

- Built 32-bit MIPS processor in SystemVerilog; added JAL/JR with 3-way register MUX and 4-way PC source MUX for subroutine support
- Verified with assembly subroutine test program; optimized critical path from 42.6 MHz to 52.3 MHz via TimeQuest

GCD Hardware Implementation

Fall 2024

- Implemented Euclidean GCD in hardware using 6-state FSM on DE2-115; switch inputs, debounced start, 7-segment output
- 122 MHz max frequency, +11.8 ns slack, only 87 logic elements (0.08%); verified across 1,000 random test cases

PROFESSIONAL EXPERIENCE

Construction & Electrical Specialist | Lazcano Family Construction

2019 – Present

- Lead NEC-compliant electrical installations: 200A panel upgrades, branch circuit replacement, GFCI/AFCI across 50+ projects; 100% inspection pass rate
- 8,000+ sq ft hardwood flooring, 8 full bathroom renovations — apply circuit analysis and load calculations on every job

Team Member | Target Corporation

May 2023 – Present

- High-volume retail customer service; bilingual Spanish/English

Lifeguard | MGM Resorts International

2020 – 2022

- Monitored guest safety at high-traffic resort; maintained certifications, responded to emergencies

Portfolio & project documentation: will-l10.github.io